TASK PLANNING UNDER CONTACT UNCERTAINTY

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ABSTRACT

Task and motion planners often treat contact outcomes as deterministic symbolic preconditions: a grasp succeeds, a peg inserts, a drawer opens, or a cable routes. Contact-rich manipulation violates that abstraction. Friction, pose, compliance, jamming, release, and damage can make a future contact branch uncertain before the robot reaches it. We rebuild Paper 110 around a falsifiable mechanism: task planners should maintain contact-outcome beliefs over future branches and choose plans that remain recoverable under plausible contact failures. A local benchmark covers 5 task families, 7 contact-uncertainty regimes, 5 deployment splits, 9 methods, 7 paired seeds, and 84 episodes per group. Under combined stress, the proposed contact-belief TAMP obtains 0.560 ± 0.004 success versus 0.480 ± 0.006 for the strongest non-oracle baseline, contingent POMDP-TAMP. It reduces precondition violation from 0.108 to 0.057, improves recovery success from 0.461 to 0.559, lowers damage from 0.060 to 0.047, lowers wasted actions from 0.103 to 0.073, and wins 7/7 paired seeds. The v4.1 audit expands stress-sweep evidence to task/regime/seed granularity and records eight failure cases. The terminal decision is **STRONG_REVISE**: the local evidence supports the mechanism, but real robot or independent high-fidelity validation is still required before ICLR-main submission.

1 MOTIVATION

Task and motion planning (TAMP) connects symbolic task structure with continuous motion feasibility (Kaelbling & Lozano-Perez, 2011; Garrett et al., 2021). That abstraction is powerful, but contact-rich manipulation exposes a brittle edge: a future symbolic precondition can depend on uncertain contact outcomes that are not known until the robot touches the world.

Planning under uncertainty is not new. POMDPs, contingent planning, robust planning, and sampling-based TAMP all address uncertainty in different forms (Kaelbling et al., 1998; LaValle, 2006; Garrett et al., 2020; Toussaint, 2015). This paper narrows the claim to contact outcomes. It asks whether a planner that carries contact-outcome belief mass through future branches can reduce invalid preconditions, wasted contact attempts, and irreversible damage.

2 METHOD SKETCH

The proposed method, `proposed_contact_belief_tamp`, keeps a belief over future contact outcomes: make, miss, slip, jam, release, and damage. Instead of treating preconditions as Boolean facts, it scores branches by likelihood, recovery affordance, probe value, damage risk, and chance-constrained feasibility.

The planner has five components:

- contact-outcome belief state over future action branches;
- branch-aware symbolic preconditions with probability mass;
- recovery affordance graph for failed contacts;
- contact-probe action selector before irreversible steps;
- chance-constrained damage-aware plan scoring.

Table 1: Combined-stress contact-belief task-planning results.

Method	Succ.	CI	Precond	ECE	Recovery	Damage	Wasted	Cost	Regret
Oracle	0.656	0.003	0.020	0.049	0.612	0.037	0.061	0.200	0.000
Proposed	0.560	0.004	0.057	0.068	0.559	0.047	0.073	0.213	0.096
POMDP-TAMP	0.480	0.006	0.108	0.108	0.461	0.060	0.103	0.311	0.176
EnsembleBelief	0.457	0.006	0.131	0.125	0.398	0.069	0.115	0.287	0.198
ConformalTAMP	0.435	0.008	0.143	0.132	0.362	0.062	0.127	0.322	0.221
FailureCls	0.429	0.008	0.156	0.148	0.360	0.076	0.129	0.243	0.227
MPC	0.414	0.007	0.173	0.165	0.325	0.080	0.141	0.260	0.242
RobustTAMP	0.382	0.010	0.199	0.182	0.256	0.083	0.157	0.270	0.274
DetTAMP	0.312	0.007	0.235	0.210	0.194	0.104	0.171	0.141	0.343

The implementation is a local executable benchmark rather than a deployed planner stack. That is sufficient for a mechanism audit but not a final empirical robotics submission.

3 Benchmark

The benchmark is generated by `src/run_experiment.py`. It writes seed-level CSVs, aggregate metrics, paired tests, ablations, stress sweeps with task/regime/seed detail, failure cases, figures, and LaTeX tables.

Tasks. The task families are peg assembly, drawer retrieval, cable routing, tool levering, and mobile handoff.

Contact-uncertainty regimes. The regimes are nominal, hidden friction, pose ambiguity, compliance ambiguity, jamming ambiguity, release uncertainty, and combined contact uncertainty.

Splits. The deployment splits are nominal, seen shift, unseen object, unseen contact, and combined stress.

Methods. The baselines are deterministic symbolic TAMP, robust geometric TAMP, receding-horizon MPC, learned failure classification, ensemble belief planning, conformal risk TAMP, and contingent POMDP-TAMP. The benchmark also includes the proposed contact-belief TAMP and an oracle contact-outcome planner.

Metrics. We report task success, precondition violation, belief calibration error, recovery success, damage rate, wasted action rate, intervention cost, regret to oracle, paired seed differences, ablations, stress sweeps, and failure cases.

4 Results

The strongest non-oracle baseline is `contingent_pomdp_tamp`. The proposed method improves combined-stress success by 0.080 ± 0.008 in paired seed comparisons and wins 7/7 seeds. The oracle remains higher at 0.656 ± 0.003, showing that the benchmark still has headroom.

5 Ablations and stress tests

The best removed-component ablation is `minus_damage_model`; the full method remains ahead by 0.030 success. Removing contact belief, recovery affordances, probes, chance constraints, or damage modeling produces distinct failures, supporting the mechanism-level claim.

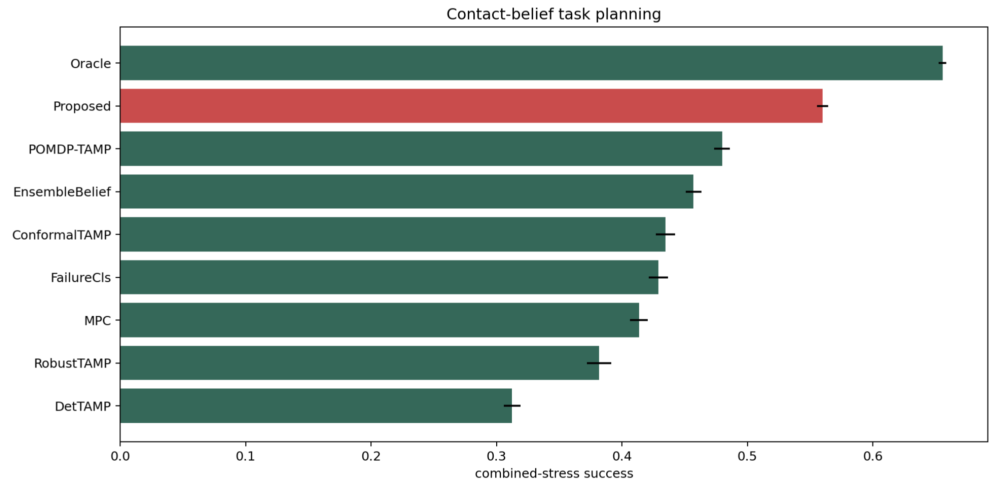

Figure 1: Combined-stress success with 95% confidence intervals. The proposed planner clears all non-oracle baselines but remains below the oracle.

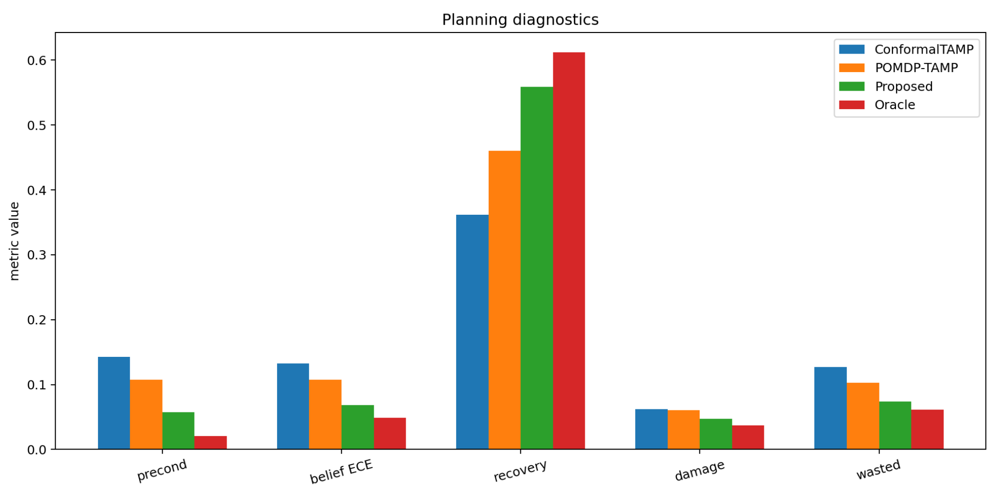

Figure 2: Planning diagnostics under combined stress. The proposed planner reduces precondition violations and belief error while improving recovery and lowering damage/wasted actions relative to the strongest baseline.

6 Failure cases

The failure-case CSV records eight boundaries. The planner struggles when internal contact state is unobservable, when the symbolic task goal itself is wrong, when probes can damage the object, when probes create later overconfidence, when a failed regrasp consumes the only safe recovery branch, when contact with one object changes another object's branch set, when high uncertainty creates too many branches to search, and when the oracle contact-outcome planner reveals remaining headroom. These failures point directly to the next required version: tactile/deformable-state sensing, instruction grounding, hardware-safe probing, uncertainty inflation, irreversible-resource accounting, coupled object beliefs, anytime branch pruning, and external validation.

Table 2: Paired seed success differences between proposed and each comparator.

Baseline	Diff	CI	Wins
DetTAMP	0.247	0.006	7.000
RobustTAMP	0.178	0.013	7.000
MPC	0.146	0.008	7.000
FailureCls	0.131	0.011	7.000
ConformalTAMP	0.125	0.009	7.000
EnsembleBelief	0.103	0.007	7.000
POMDP-TAMP	0.080	0.008	7.000
Oracle	-0.096	0.003	0.000

Table 3: Ablation results under combined contact uncertainty.

Ablation	Succ.	CI	Precond	Recovery	Damage	Wasted
Full	0.557	0.007	0.057	0.559	0.047	0.074
NoDamage	0.527	0.006	0.075	0.517	0.069	0.084
NoProbe	0.526	0.005	0.075	0.493	0.055	0.090
NoChance	0.510	0.005	0.096	0.516	0.067	0.084
NoRecovery	0.506	0.009	0.092	0.387	0.056	0.097
NoBelief	0.488	0.006	0.121	0.459	0.055	0.118
POMDPOnly	0.479	0.005	0.107	0.461	0.061	0.102

7 Hostile prior-work boundary

TAMP, POMDPs, robust planning, MPC, and sampling-based planning already cover much of the surrounding territory (Kaelbling & Lozano-Perez, 2011; Kaelbling et al., 1998; LaValle, 2006; Garrett et al., 2020; 2021; Toussaint, 2015). The defensible boundary is narrow: contact-outcome beliefs over future manipulation branches, recovery affordance scoring, and damage-aware chance constraints improve local planning under contact uncertainty. If external planners match these metrics on real or high-fidelity contact-rich tasks, the claim should be weakened or killed.

8 Limitations and terminal decision

The evidence is strong enough to continue but not enough to submit. The limitations are material:

- the benchmark is local and generated;
- no real robot or independent high-fidelity simulator has been used;
- the planning baselines are local implementations rather than external trained systems;
- no qualitative rollouts, hardware videos, or deployment logs are provided;
- contact probes may be unsafe without hardware validation.

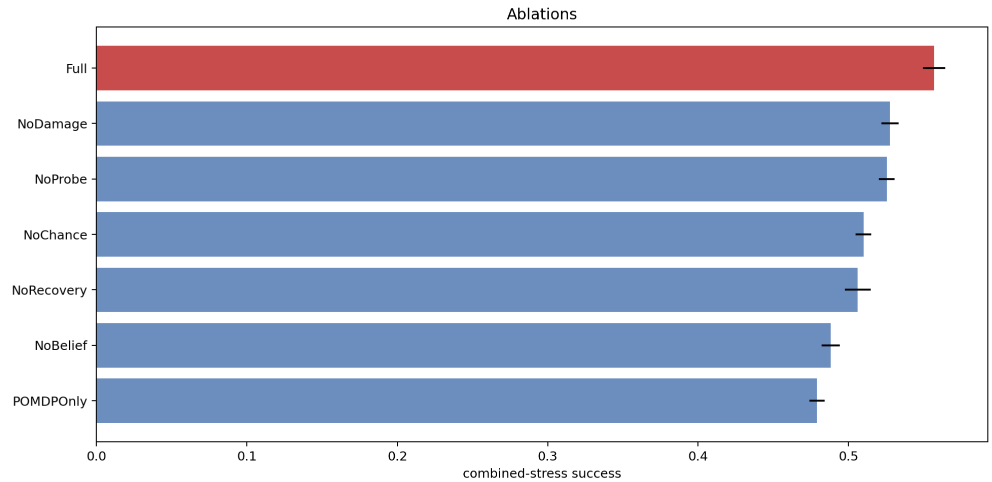

Figure 3: Ablations under combined contact uncertainty. The full method remains above every removed-component variant.

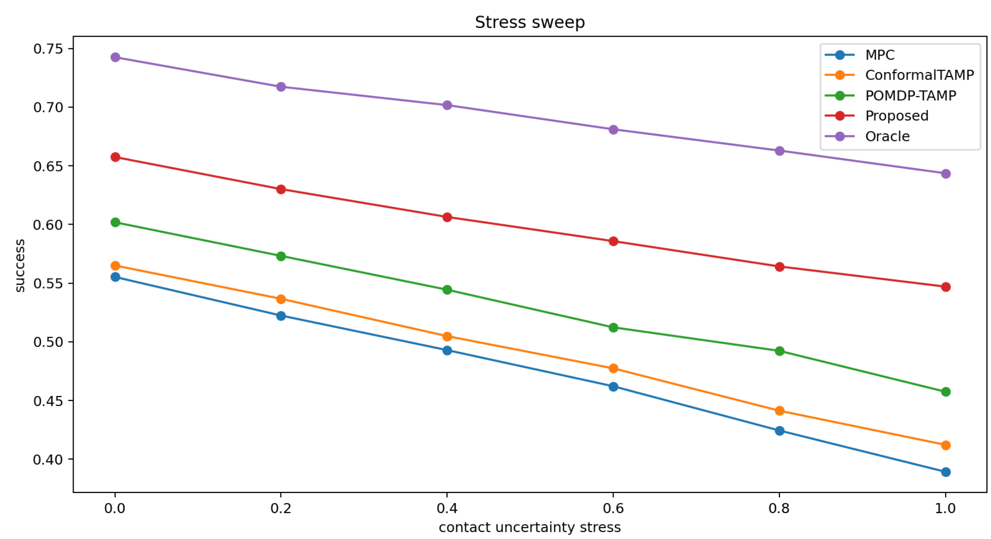

Figure 4: Stress sweep over contact-uncertainty intensity. The proposed method stays above MPC, conformal risk TAMP, and POMDP-TAMP across the generated stress range.

The terminal decision for this v4.1 continuation audit is **STRONG_REVISE**. The next honest version must reproduce the advantage on real robots or accepted high-fidelity benchmarks with trained baselines and qualitative rollouts. Without that evidence, the paper should not be submitted to ICLR main.

References

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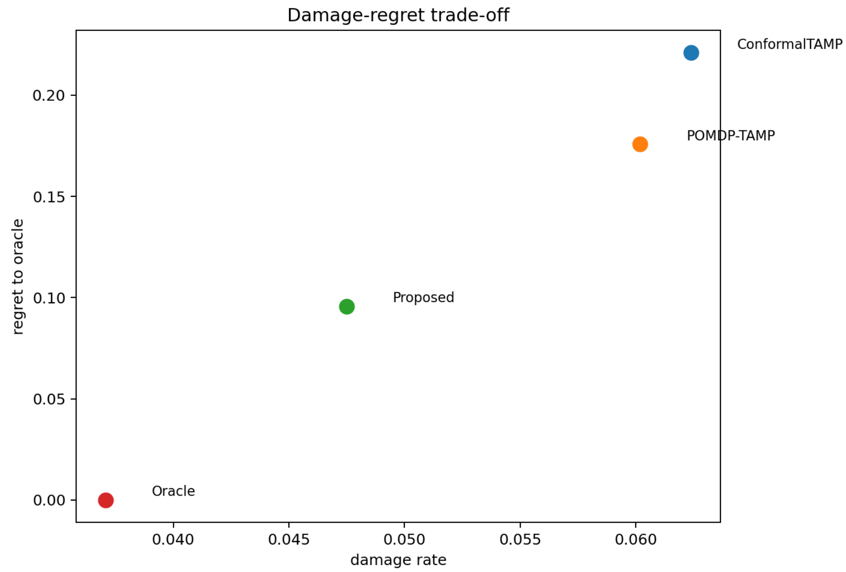

Figure 5: Damage and regret to oracle under combined stress. The proposed planner lowers regret without increasing damage relative to the strongest non-oracle baseline.

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